This guide provides a comprehensive, step-by-step solution for **Assignment 1 - Part C**, covering annuity valuations, perpetuities, and effective annual rates.

Understanding the "How" and "Why"

In finance, we use specific formulas to translate future money into today's value (**Present Value**) or current money into future value (**Future Value**).

- **Ordinary Annuity:** Payments occur at the **end** of each period (e.g., a standard loan payment).

- **Annuity Due:** Payments occur at the **beginning** of each period (e.g., rent). Because money starts earning interest one period earlier, an annuity due is always worth more than an ordinary annuity.

- **Compounding:** The process of earning "interest on interest." The more frequently it happens, the faster the money grows.

Question 1: Future Value of Annuities

a. $500 per year for 8 years at 14% (Ordinary Annuity)

- **Formula:** $FVA_{ordinary} = PMT \times \frac{(1+r)^n - 1}{r}$
- **Calculation:** $500 \times \frac{(1.14)^8 - 1}{0.14} = 500 \times 13.2328 = \mathbf{\$6,616.38}$
- **Why:** We are finding out how much eight $500 deposits will grow to if each one earns 14% interest until the end of the 8th year.

b. $250 per year for 4 years at 7% (Ordinary Annuity)

- **Calculation:** $250 \times \frac{(1.07)^4 - 1}{0.07} = 250 \times 4.4399 = \mathbf{\$1,110.00}$

c. Rework parts a and b as Annuities Due

- **Why:** For an annuity due, you multiply the ordinary annuity result by $(1+r)$ because every payment earns one extra year of interest.

- **Part A (Due):** $\$6,616.38 \times 1.14 = \mathbf{\$7,542.67}$
- **Part B (Due):** $\$1,110.00 \times 1.07 = \mathbf{\$1,187.70}$

Question 2: Present Value of Annuities

a. $600 per year for 12 years at 8% (Ordinary Annuity)

- **Formula:** $PVA_{ordinary} = PMT \times \frac{1 - (1+r)^{-n}}{r}$
- **Calculation:** $600 \times \frac{1 - (1.08)^{-12}}{0.08} = 600 \times 7.5361 = \mathbf{\$4,521.65}$
- **Why:** This calculates the lump sum you would need today to be able to withdraw $600 every year for 12 years, assuming your remaining balance earns 8% interest.

b. $300 per year for 6 years at 4% (Ordinary Annuity)

- **Calculation:** $300 \times \frac{1 - (1.04)^{-6}}{0.04} = 300 \times 5.2421 = \mathbf{\$1,572.64}$

d. Rework parts a and b as Annuities Due

- **Why:** Just like future value, the present value of an annuity due is higher because the first payment is received immediately (Day 0) and doesn't need to be discounted.

- **Part A (Due):** $\$4,521.65 \times 1.08 = \mathbf{\$4,883.38}$
- **Part B (Due):** $\$1,572.64 \times 1.04 = \mathbf{\$1,635.55}$

Question 3: Perpetuities

Problem: What is the PV of a $600 perpetuity at 5%? What if the rate doubles to 10%?

- **Formula:** $PV = \frac{PMT}{r}$
- **At 5%:** $\frac{600}{0.05} = \mathbf{\$12,000}$
- **At 10%:** $\frac{600}{0.10} = \mathbf{\$6,000}$
- **Why:** A perpetuity is a payment that lasts forever. There is an **inverse relationship** between interest rates and present value; when the interest rate doubles, the value of the investment is cut in half.

Question 4: Effective Annual Rate (EAR)

Problem: Bank A pays 2% compounded annually. Bank B pays 1.75% compounded daily.

a. Which bank should you use based on EAR?

- **Formula:** $EAR = (1 + \frac{i_{nom}}{n})^n - 1$
- **Bank A:** Since it is annual, the $EAR = \mathbf{2.00\%}$

- **Bank B:** $(1 + \frac{0.0175}{365})^{365} - 1 = \mathbf{1.765\%}$

- **Decision:** Choose **Bank A**. Even though Bank B compounds daily, its base interest rate is too low to beat Bank A's 2% rate.

b. Does the timing of withdrawals matter?

- **Yes.**

- **Explanation:** If you must leave funds for an "entire compounding period" to earn interest, Bank A is risky. If you withdraw your money after 11 months, you earn **0% interest** at Bank A. At Bank B, you earn interest every single day, so a mid-year withdrawal still yields a return. If you need flexibility, Bank B is better; if you are certain you can wait a full year, Bank A is better.